Star Milvus for a chance to win a prize tonight!

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This just dropped (open-source edition)!

https://github.com/canopyai/Orpheus-TTS

https://huggingface.co/ds4sd/SmolDocling-256M-preview

https://huggingface.co/nvidia/GR00T-N1-2B

The world is much more than just text and keywords

90% of newly generated data in 2025 will be unstructured data

*Data Source: The Digitization of the World by *IDC*

zilliz

Mission:

Helping organizations make sense of unstructured data.

2017
Founded

$113M
Raised

140+
Employees

Redwood City, CA
Headquarters

zilliz

Welcome Speakers

TECH TALK 1

From O1 to O3: How Reasoning Transformed AI and Launched a New Generation of Agents

Harry Zhao
Member of Technical Staff, OpenAI

TECH TALK 2

Building an Agent to Reason about Private Data with OpenAI & Milvus

Stefan Webb
Developer Advocate, Zilliz

 |

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Building a Reasoning Agent with Milvus and OpenAI

Stefan Webb

Developer Advocate, Zilliz

stefan.webb@zilliz.com
https://www.linkedin.com/in/stefan-webb

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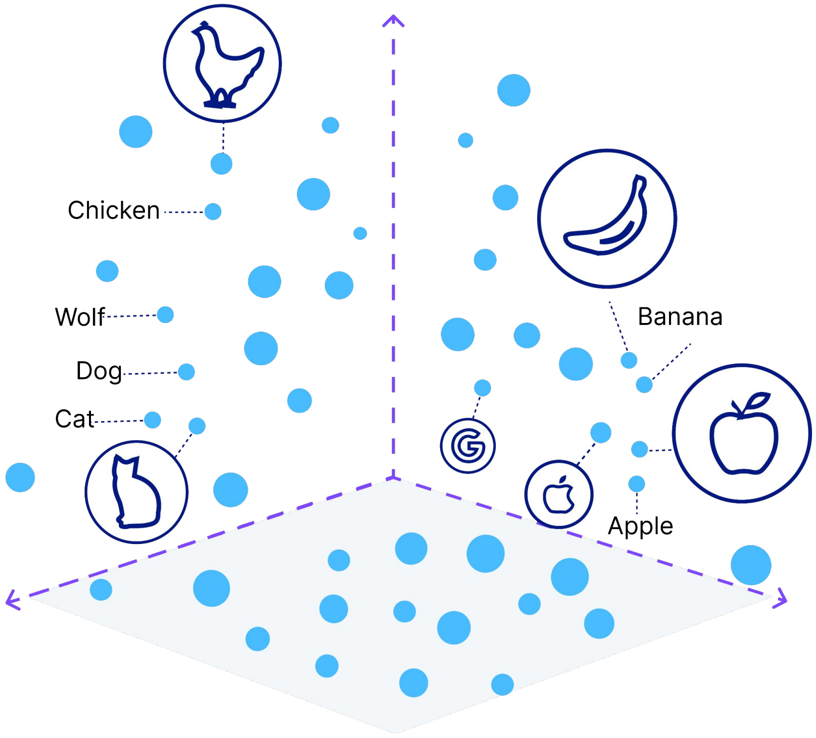
01

Why Milvus?

Vector search is the new standard



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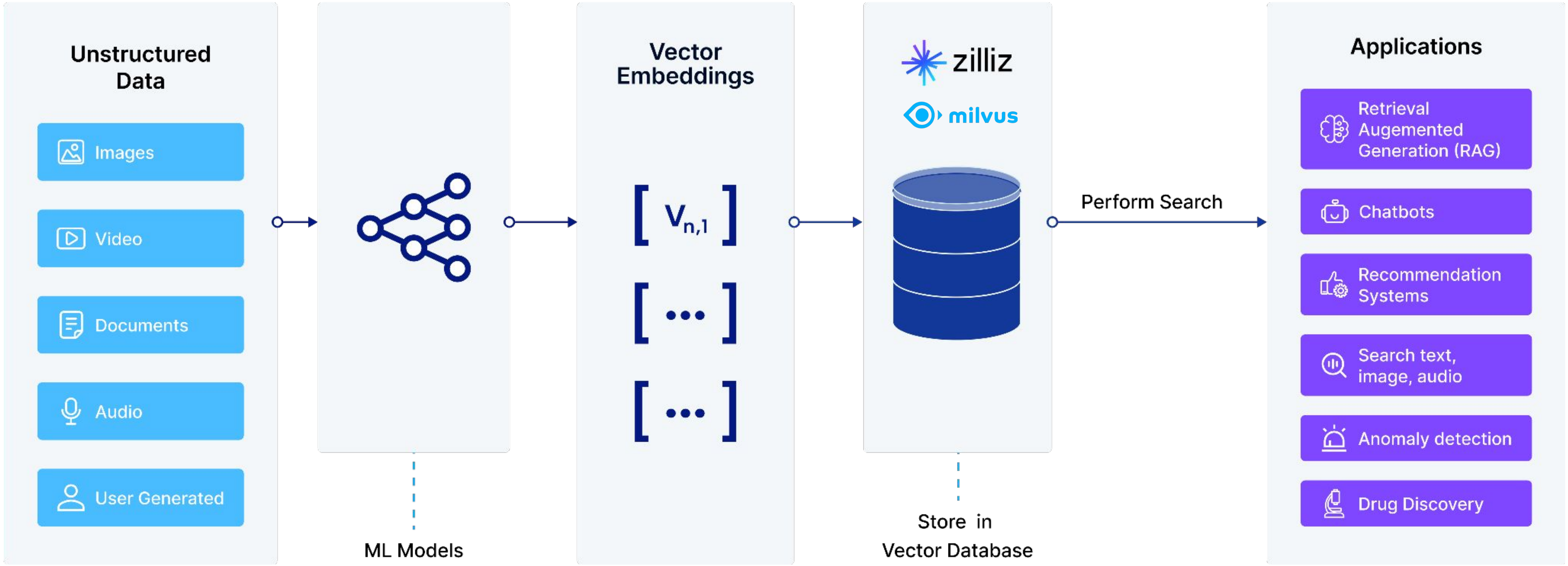
A New tool emerged. The Vector Database



 |

Milvus: The most widely-adopted vector database

Milvus is an **Open-Source Vector Database** to **store**, **index**, **manage,** and **use** the massive number of **embedding vectors** generated by deep neural networks and LLMs.

milvus
LF AI & DATA

430+	33K+	67M+	2.8K +
contributors	stars	docker pulls	forks

zilliz

Built by database & AI experts


Milvus: A Purpose-Built Vector Data Management System


Manu: A Cloud Native Vector Database Management System

Zilliz was built by a top-tier team of **algorithm and database engineers** with a strong pedigree in developing **high-performance**, **scalable**, and **highly available** distributed systems, uniquely tailored for **vector search.**

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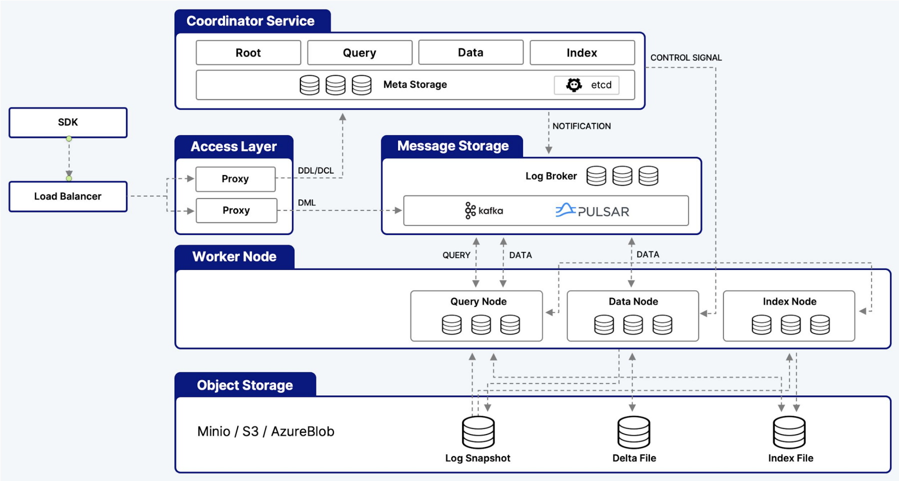
Milvus Architecture

Design Principles

- Separation of storage and compute
- Fully depend on mature storage systems
- Microservice - scale by functionality
- Separate streaming and historical data
- Pluggable engine, storage and index
- Log as data

Fully distributed, designed for scalability

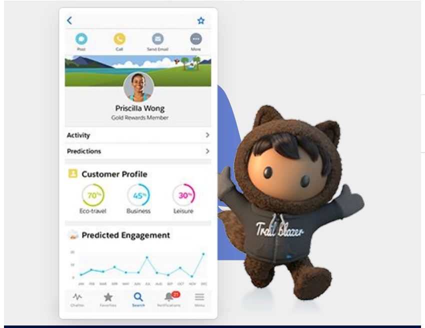


Rich functionality

- Dynamic Schema
- Float, Binary, & Sparse Vector
- Tag+Vector Optimized Filtering
- Hybrid Search Dense & Sparse
- RBAC, TLS, Encryption
- Million+ level tenant support
- Disk Based Index
- Tiered Storage
- Bulk Import
- GPU, Intel & ARM CPU support

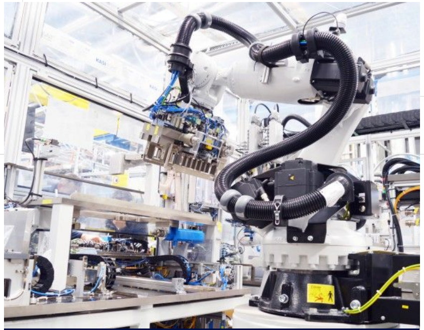
Industry leaders already use vector search in their apps

Use Case: Data Search	**Use Case**: Drug Discovery	**Use Case**: Image Search	**Use Case:** Recommender System
Vectors: 2 Billion	**Vectors:** 12 Billion	**Vectors**: 20 Billion	**Vectors**: 20 Billion
Req'ts: 200 ms, Cost mgmt	**Req'ts:** High Recall	**Req'ts**: High Insertion, Cost	**Req'ts**: 5,000 QPS
Index: DiskANN for cost savings	**Index:** BIN_FLAT	**Index**: Disk Based Index	**Index**: HNSW & CAGRA



salesforce

REGENERON



BOSCH

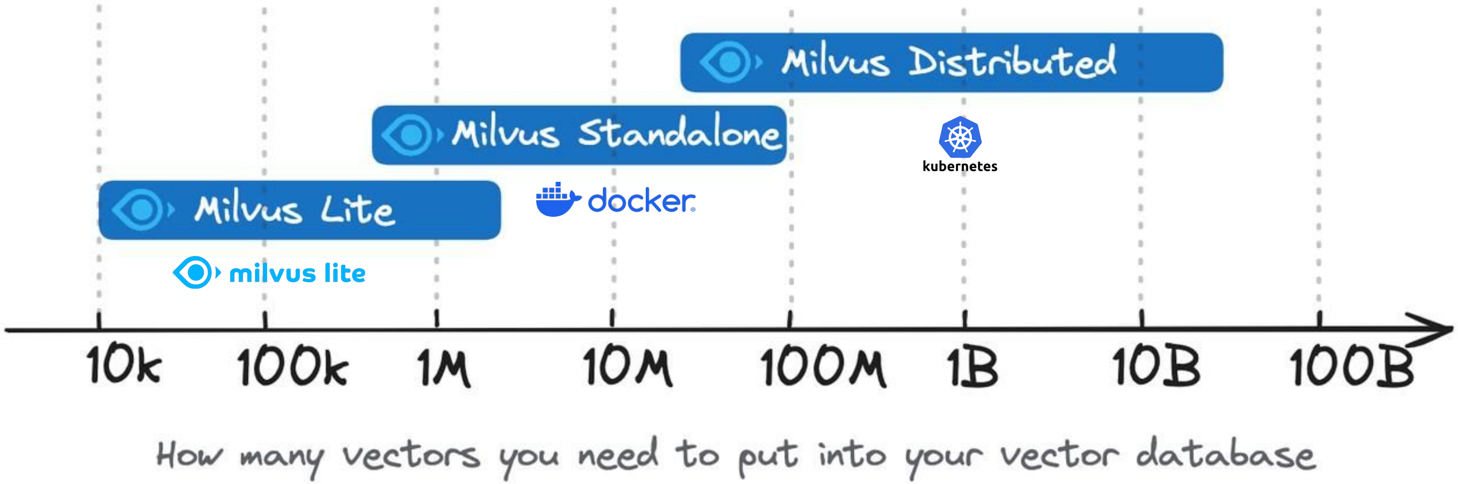
SmartNews

zilliz

Milvus Users

zilliz

Flexible Deployment Options for Milvus



milvus

Set up Once: Common API across all products regardless of architecture

 |

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AI Stack

Framework

LangChain, Semantic Kernel, LlamaIndex, haystack by deepset

Software Infrastructure

Embedding Models

PyTorch, Hugging Face, cohere

Vector Database

milvus, zilliz

LLMs

OpenAI, Inflection, LLaMA by Meta, ANTHROPIC

Hardware Infrastructure

NVIDIA, intel, aws, Google Cloud, Azure

zilliz

02

DeepSearcher demo

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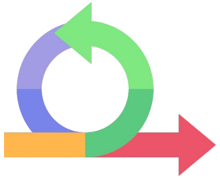
Research Agents



Iteration

"…learned to plan and execute a multi-step trajectory…"

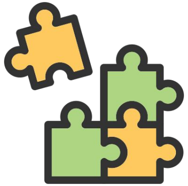
"…backtracking and reacting to real-time information…"

"…pivoting as needed in reaction to information it encounters…"

Search

"…trained using end-to-end reinforcement learning on hard browsing and reasoning tasks across a range of domains…"

"…optimized for web browsing and data analysis…"

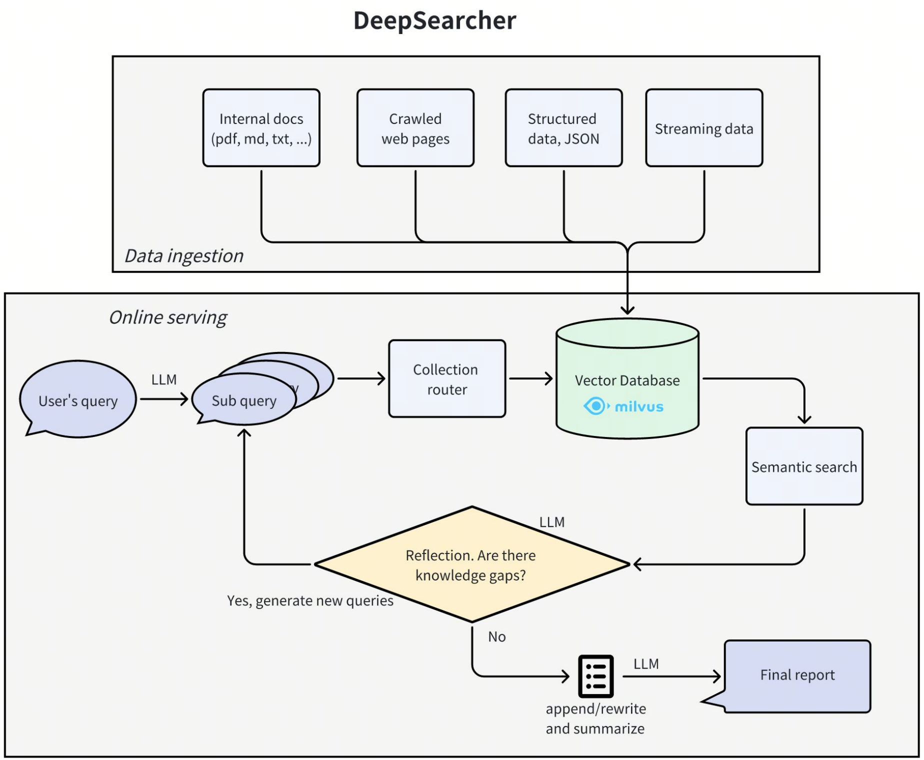


Reasoning

"…fine-tuned on the upcoming OpenAI o3 reasoning model…"

"""…leverages reasoning to search, interpret, and analyze massive amounts of text…"""

DeepSearcher from Zilliz



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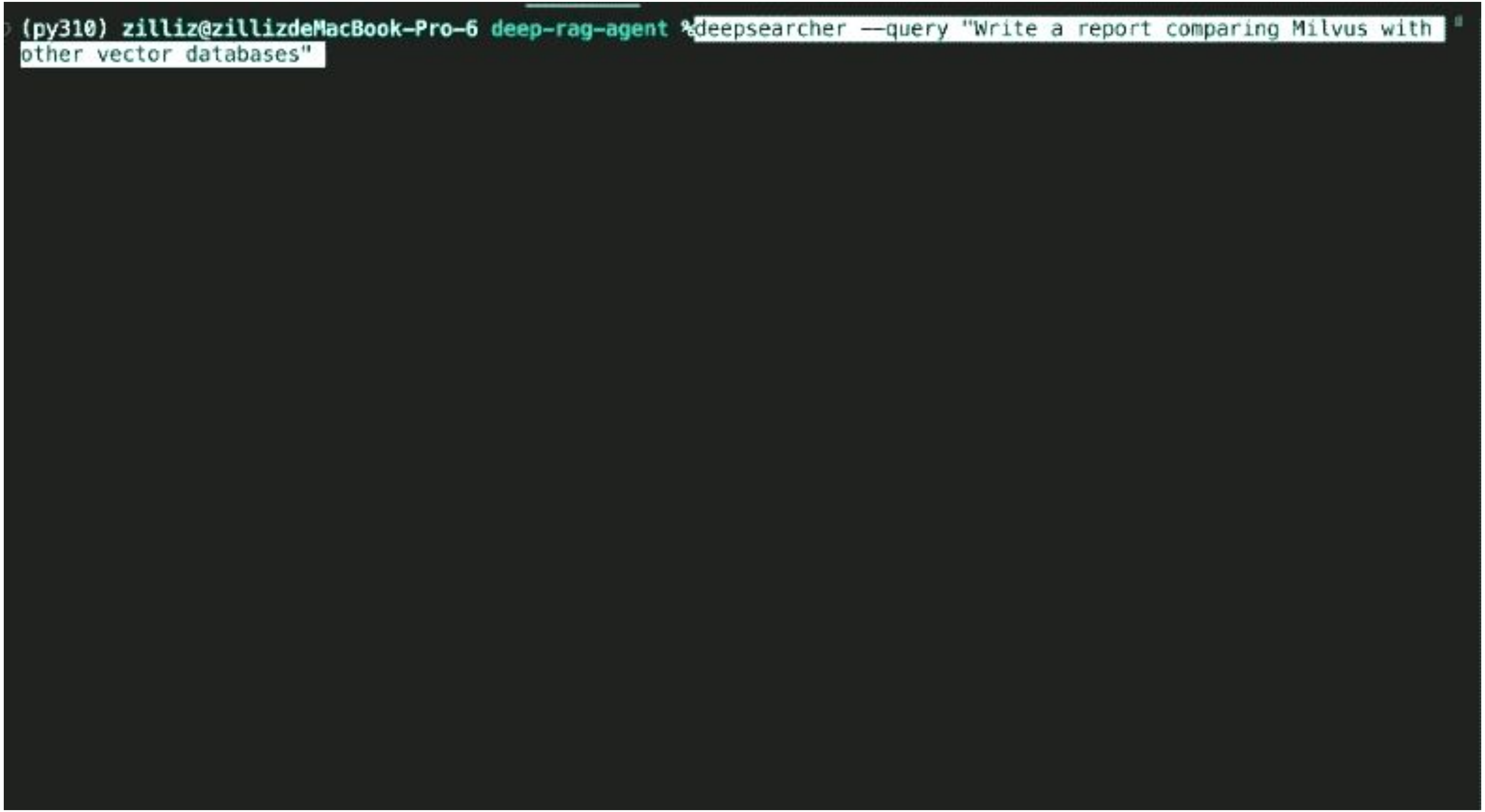
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher --query "Write a report comparing Milvus with other vector databases"


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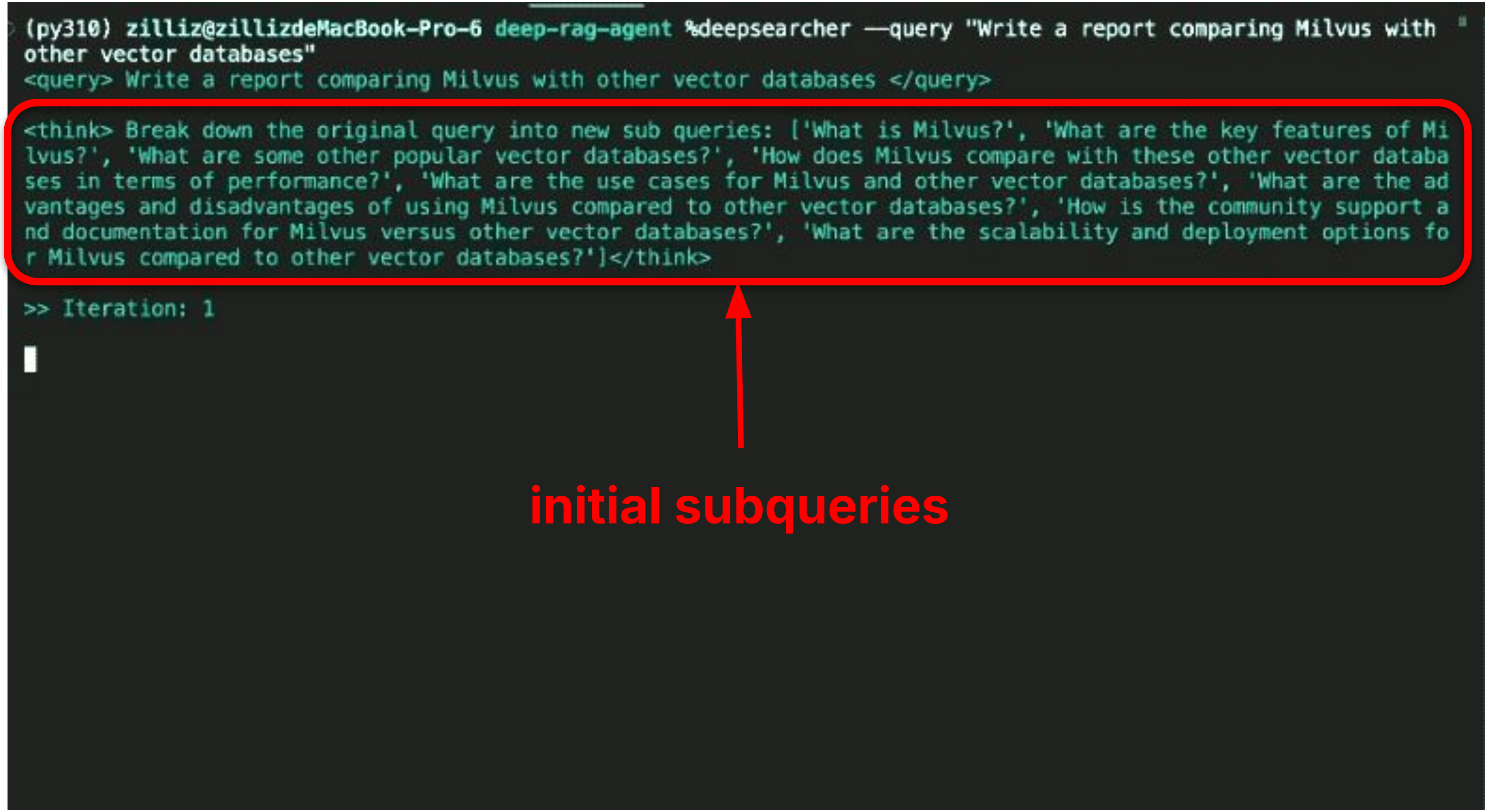
```

our query/task

zilliz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<think> Break down the original query into new sub queries: ['What is Milvus?', 'What are the key features of Milvus?', 'What are some other popular vector databases?', 'How does Milvus compare with these other vector databases in terms of performance?', 'What are the use cases for Milvus and other vector databases?', 'What are the advantages and disadvantages of using Milvus compared to other vector databases?', 'How is the community support and documentation for Milvus versus other vector databases?', 'What are the scalability and deployment options for Milvus compared to other vector databases?']</think>
>> Iteration: 1
initial subqueries

```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher —query "Write a report comparing Milvus with other vector databases"
<query> Write a report comparing Milvus with other vector databases </query>

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>> Iteration: 1

<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think>

<search> Search [What is Milvus?] in [milvus_docs]...  </search>
```

think what action to perform
search 1st subquery, 1st source

zilliz

```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher —query "Write a report comparing Milvus with other vector databases"
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<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think>

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<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integrations/apify_milvus_rag.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/roadmap.md'] </search>

<search> Search [What is Milvus?] in [what_is_milvus]...  </search>
```

retrieve chunks with Milvus

zilliz

```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher —query "Write a report comparing Milvus with other vector databases"
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<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think>

<search> Search [What is Milvus?] in [milvus_docs]... </search>

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<search> Search [What is Milvus?] in [what_is_milvus]...  </search>

<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatisMilvus.pdf', './examples/data/WhatisMilvus.pdf', './examples/data/WhatisMilvus.pdf'] </search>
```

search 1st subquery, 2nd source

retrieve chunks with Milvus

zilliz

```
<think> Break down the original query into new sub queries: ['What is Milvus?', 'What are the key features of Mi
lvus?', 'What are some other popular vector databases?', 'How does Milvus compare with these other vector databa
ses in terms of performance?', 'What are the use cases for Milvus and other vector databases?', 'What are the ad
vantages and disadvantages of using Milvus compared to other vector databases?', 'How is the community support a
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r Milvus compared to other vector databases?']</think>

>> Iteration: 1

<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think
>

<search> Search [What is Milvus?] in [milvus_docs]...  </search>

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<search> Search [What is Milvus?] in [what_is_milvus]...  </search>

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sMilvus.pdf', './examples/data/WhatisMilvus.pdf'] </search>

<think> Perform search [What are the key features of Milvus?] on the vector DB collections: ['milvus_docs', 'mil
vus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What are the key features of Milvus?] in [milvus_docs]...  </search>
```

think what action to perform

search 2nd subquery, 1st source

zilliz

```
>> Iteration: 1

<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think
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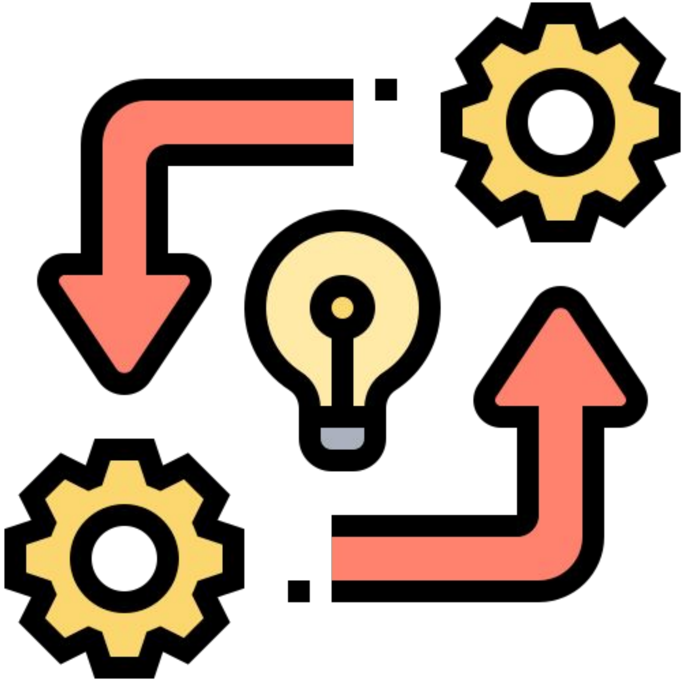
```

retrieve chunks with Milvus

zilliz



Accelerated Playback

reflect on iter 1, new subqueries

```
<search> Search [What are the scalability and deployment options for Milvus compared to other vector databases?]
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etstarted/install-overview.md'] </search>

<think> Reflecting on the search results... </think>

<think> New search queries for next iteration: ['What are the performance benchmarks of Milvus compared to its c
ompetitors?', 'What specific use cases are best suited for other vector databases compared to Milvus?', 'What ar
e the recent developments and future directions in vector databases beyond Milvus?'] </think>

>> Iteration: 2

<think> Perform search [What are the performance benchmarks of Milvus compared to its competitors?] on the vecto
r DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_docs]...
  </search>
```


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Accelerated Playback

search 1st subquery

```
<search> Search [What are the scalability and deployment options for Milvus compared to other vector databases?]
 in [milvus_docs]...  </search>

<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integra
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```
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Accelerated Playback

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x/site/en/reference/architecture/architecture_overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/g
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  </search>

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...  </search>

<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatisMilvus.pdf', './examples/data/Whati
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Accelerated Playback

Accelerated Playback

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search 2nd subquery

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Accelerated Playback

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Accelerated Playback

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Accelerated Playback

search 3rd subquery

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Accelerated Playback

Accelerated Playback

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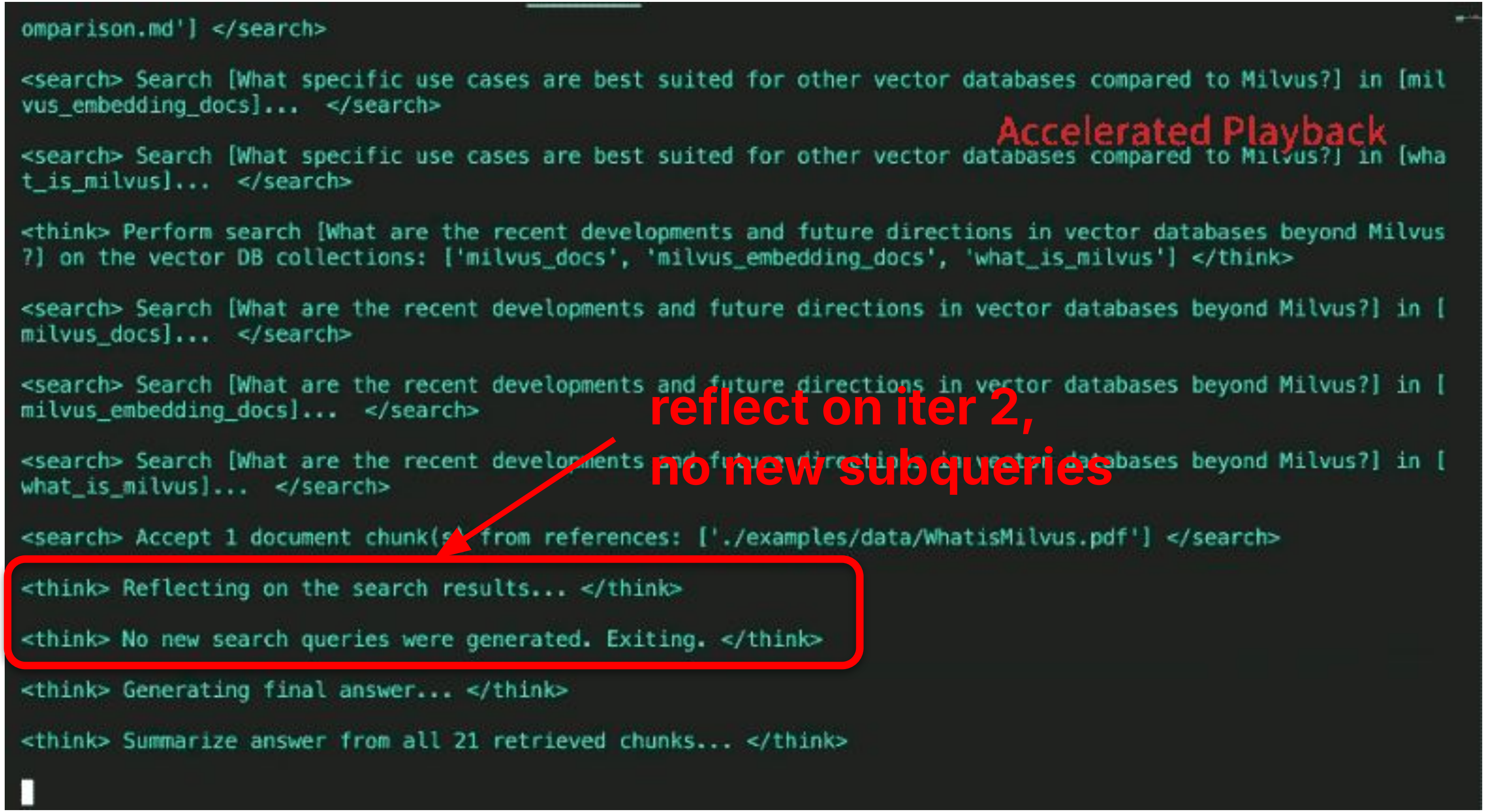
zilliz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reflect on iter 2,
no new subqueries
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<think> No new search queries were generated. Exiting. </think>
<think> Generating final answer... </think>
<think> Summarize answer from all 21 retrieved chunks... </think>

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<think> No new search queries were generated. Exiting. </think>

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==== FINAL ANSWER====

### Report: Comparison of Milvus with Other Vector Databases

**Introduction to Milvus:**
Milvus is a high-performance, scalable vector database designed for managing unstructured data like images, text
s, and audio by converting them into numerical vectors. It is open-source and available as a cloud service, supp
orting various deployment options: Milvus Lite, Standalone, and Distributed. These variants cater to different s
cales, from local testing environments to large Kubernetes deployments capable of handling tens of billions of v
ectors.

**Key Features of Milvus:**
1. **High Performance**: Milvus is optimized for speed and efficiency. It employs advanced data access patterns
and hardware-aware optimizations, achieving a performance boost of 30%-70% over popular implementations like FAI
SS and HNSWLib.
2. **Flexible Deployment**: It supports various deployment options—Milvus Lite for lightweight applications, Sta
ndalone for single-host setups, and Distributed architecture for high-scale scenarios.
3. **Robust Indexing Algorithms**: Milvus supports diverse indexing/search algorithms (IVF, HNSW, DiskANN) enhan
cing query speed through real-time processing.
4. **Advanced Querying Capabilities**: The database supports multiple types of searches, including ANN, hybrid,
and filtered searches, alongside maintaining support for advanced data types like sparse and binary vectors.

**Comparative Analysis:**
```

generate report

```
2. **Flexible Deployment**: It supports various deployment options—Milvus Lite for lightweight applications, Sta
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and filtered searches, alongside maintaining support for advanced data types like sparse and binary vectors.

**Comparative Analysis:**
1. **Performance**: Milvus often outperforms its competitors (like Pinecone and others) by a factor of 2-5x due
to its efficient engine implemented in C++, optimized for low-level resource management and multi-threading.
2. **Scalability**: With its cloud-native architecture, Milvus scales linearly to handle significant loads and s
upports concurrent queries efficiently. It managed billion-scale vectors with minimal performance degradation du
ring extensive testing.
3. **Community Support and Documentation**: Milvus has robust community engagement and extensive documentation c
ompared to many other vector databases. Its integrations and user contributions, especially from large corporati
ons, enhance its reliability and development support.
4. **Use Cases**: Milvus excels in applications needing intense similarity searches, such as image retrieval, AI
-driven applications, and machine learning frameworks. However, some scenarios may better suit specialized datab
ases like Pinecone for specific niche applications.

**Advantages and Disadvantages:**
- **Advantages**:
   - Superior performance and speed.
   - Flexible regarding deployment options.
   - Comprehensive SDK support and community-driven documentation.
- **Disadvantages**:
   - Complexity in setup for new users.
   - Requires careful optimization for specific workloads.

**Recent Developments and Future Directions:**
Milvus is continuously evolving with planned updates that focus on enhancing its speed, adopting more advanced
```

```
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  - Requires careful optimization for specific workloads.

**Recent Developments and Future Directions:**
Milvus is continuously evolving with planned updates that focus on enhancing its speed, adopting more advanced i
ndexing mechanisms, and exploring serverless architectures for even greater scalability and cost-effectiveness.

**Conclusion:**
Milvus is a compelling choice among vector databases, noteworthy for its high performance, flexible deployment o
ptions, and strong community support. While it shines in many applications, specific use cases might warrant loo
king into tailored alternatives depending on organizational needs and operational scale.
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %
```


zilliz

03

Let's build it ourselves!

zilliz

It's Coding Time!

https://bit.ly/41HjZbJ

04

Where to from here?

zilliz

Where to from here?

Data

- Open-source
- High-quality
- Reasoning traces

Cost

- Specialized hardware
- Algorithmic improvements

Scaling

- Per-se

Zilliz Offerings

https://cloud.zilliz.com/signup

SELF MANAGED SOFTWARE	FULLY MANAGED SERVICE	BRING YOUR OWN CLOUD
Milvus	**Zilliz Cloud**	**Zilliz BYOC**
Most widely-adopted open source vector database	AI Powered Search that is performant and scales	For Private VPCs
milvus lite, docker, kubernetes	aws, Google Cloud, Azure	aws, Google Cloud, Azure *Coming Soon!*

milvus

Set up Once: Common API across all products regardless of architecture

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LET'S STAY CONNECTED!

Stefan Webb

Developer Advocate, Zilliz

https://milvus.io/discord

Book a Free 1:1 Session

For Support Productionizing Milvus

http://bit.ly/43YkEYW

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